Loan Default Prediction — Project Report

Abstract

This project prepares a loan dataset for predicting whether a customer will default on a loan. The work completed so far covers data exploration, cleaning and feature preparation on 32,586 records; model training and evaluation are not yet in the notebook.

The raw data has clear quality problems: impossible ages and employment durations, currency symbols inside numeric fields, six duplicate rows, and missing values in four predictors. Historical default status is the worst, with about 64% of values missing. The pipeline removes duplicates and incomplete records, converts currency fields to numbers, replaces extreme values with the median, scales numeric features with a robust scaler, and encodes the categorical features.

The output is a model-ready feature table. Section 5 lists several cleaning steps that need correcting before modelling begins.

1. Introduction and objectives

Lenders lose money when borrowers default, so estimating default risk before or during a loan is a core credit-risk task. This project frames it as a binary classification problem: given a customer's profile and loan terms, predict the loan's current status, either `DEFAULT` or `NO DEFAULT`.

The objectives are:

- Understand the structure, quality and distributions of the loan dataset.
- Clean the data: remove duplicates and incomplete records, fix malformed fields, and handle impossible values.
- Transform the features into a numeric form that machine-learning models can use.
- Train and compare classification models on the prepared data (not yet done).

The analysis is written in Python in a Jupyter notebook, using pandas and NumPy for data handling, seaborn and matplotlib for plots, and scikit-learn for preprocessing.

2. Dataset

The dataset (`LoanDataset.csv`) has 32,586 records and 13 columns: 12 describe the customer and the loan, and `Current_loan_status` is the target. Money values are in pounds (£). Ranges below are from the raw data, before cleaning.

Column	Type	Description / observed values	Missing (raw)
`customer_id`	Identifier	Customer number; not a predictive feature	3
`customer_age`	Integer	Age; raw range 3 to 144	0
`customer_income`	Currency (£)	Stored as text; maximum £6,000,000	0
`home_ownership`	Categorical	RENT, OWN, MORTGAGE (seen in samples)	0
`employment_duration`	Numeric	Raw range 0 to 123; assumed to be years	895
`loan_intent`	Categorical	PERSONAL, EDUCATION, MEDICAL, VENTURE, HOMEIMPROVEMENT, DEBTCONSOLIDATION	0
`loan_grade`	Ordinal	Grade A to E	0
`loan_amnt`	Currency (£)	Stored as text, e.g. "£35,000.00"	1
`loan_int_rate`	Numeric (%)	Raw range 5.42 to 23.22	3,116
`term_years`	Integer	Loan term; range 1 to 10	0
`historical_default`	Binary (Y/N)	Whether the customer has defaulted before	20,737
`cred_hist_length`	Integer	Credit history length; range 2 to 30	0
`Current_loan_status`	Target	`DEFAULT` or `NO DEFAULT`	4

Six rows are exact duplicates. The notebook has not yet computed the share of `DEFAULT` records, so the class balance is unknown.

3. Methodology

The pipeline runs in seven stages, from first inspection of the raw file to a model-ready feature table.

1. **Exploration.** Checked the shape, a random sample, data types, null counts per column, summary statistics, duplicate rows and the correlation matrix of numeric columns.
2. **Removing incomplete records.** Dropped rows missing `customer_id`, `loan_amnt` or `Current_loan_status` (8 rows), then removed the 6 duplicate rows. This left 32,572 records.
3. **Fixing currency fields.** Stripped the £ symbol and thousands separators from `loan_amnt` and `customer_income`, then converted both to floating-point numbers.
4. **Outlier detection.** Drew box plots for interest rate, employment duration, loan amount, age, income and term, then listed rows with age above 80 and employment duration above 50.
5. **Outlier handling.** Dropped two rows, replaced eight implausible ages (84 to 144) with a median value, and replaced two employment durations of 123 with the column median.
6. **Scaling.** Applied scikit-learn's `RobustScaler` to seven numeric columns: age, income, employment duration, loan amount, interest rate, term and credit history length. It centres on the median and scales by the interquartile range, so extreme values distort it less than standard scaling.
7. **Encoding.** Mapped `loan_grade` to an ordinal scale (A=0 to E=4), `historical_default` to 1 for Y and 0 for N, and one-hot encoded `home_ownership` and `loan_intent` with `pd.get_dummies`.

The results are collected in a new table, `loan_dataset`, holding the customer ID, the seven scaled features and the encoded columns. A statistical method for filling missing interest rates was started (comparing interest rate against historical default and employment duration) but is commented out, so those gaps remain.

4. Results so far

The notebook produces a cleaned, scaled and encoded feature table, but it contains no trained model and no performance metrics yet. The table below summarises what the data inspection found and where each problem stands.

Issue	Evidence	Treatment	Status
Missing key fields	8 rows lacked `customer_id`, `loan_amnt` or `Current_loan_status`	Rows dropped	Done
Duplicate rows	6 exact duplicates	Rows dropped	Done
Currency text in numeric fields	£ symbols and commas in `loan_amnt` and `customer_income`	Converted to float	Done
Impossible ages	8 customers aged 84 to 144; minimum age in the file is 3	The 8 ages replaced with a median	Needs correcting (Section 5)
Implausible employment duration	2 rows with a value of 123	Replaced with the column median	Done
Extreme income	Maximum income £6,000,000	Intended row removal did not take effect (Section 5)	Open
Missing interest rate	3,116 rows (9.6%)	Imputation drafted, not applied	Open
Missing employment duration	895 rows (2.7%)	None	Open
Missing historical default	20,734 of 32,572 rows (about 64%) after cleaning	Left missing after encoding	Open

Two correlation results matter for modelling. Age and credit history length move together strongly (0.86), so they carry overlapping information. Interest rate has weak linear correlation with every other numeric column (between -0.06 and 0.04), which limits what a simple regression-based imputation can recover.

5. Limitations and corrections needed

Several steps in the current notebook would affect model quality or give misleading results. They should be fixed before training.

1. **Wrong column used to impute age.** The eight implausible ages are filled with the median of `employment_duration` (about 4), not of `customer_age`. Those customers now have an age of roughly 4. Use the age median or drop the rows. Ages below about 18 (the minimum is 3) were not checked.
2. **Row drop by position, not label.** `loan_history.index[[32302, 17839]]` selects by position, which no longer matches the labels after earlier rows were removed. The row labelled 32302 (age 144, income £6,000,000) still appears in later output. If the intent was to drop those labels, use `loan_history.drop(index=[32302, 17839])`.
3. **Scaler fitted on all data.** `RobustScaler` is fitted before any train/test split, so test-set statistics leak into training. Fit it on the training split only, ideally inside a scikit-learn `Pipeline`.
4. **Identifier kept with the features.** `customer_id` sits in `loan_dataset` and correlates with age (0.73) and credit history length (0.77), probably through row order. Exclude it from model inputs.
5. **Target not prepared.** `Current_loan_status` is not encoded in `loan_dataset`, and the share of defaults has not been checked. Class imbalance would change which metrics and models are appropriate.
6. **Large gap in `historical_default`.** About 64% of values are missing. The notebook has not decided whether missing means "no prior record" or "unknown", and that choice can change results. Dropping those rows would discard most of the data.
7. **Silent gaps from the grade mapping.** The `loan_grade` mapping covers A to E only. Any other grade becomes missing without warning, so check for nulls after mapping.

6. Conclusion and next steps

The data preparation stage is partly complete. The notebook has profiled a 32,586-record loan dataset, removed duplicates and incomplete rows, fixed currency fields, scaled seven numeric features and encoded the categorical ones. Model building has not started, and the corrections in Section 5 must come first, because they change the data the models will see.

Proposed next steps, in order:

1. Apply the corrections in Section 5 and re-run the notebook from the top.
2. Decide how to treat the missing values in `historical_default`, `loan_int_rate` and `employment_duration`.
3. Encode the target, check the share of defaults, and split the data into training and test sets before scaling.
4. Train a simple baseline model, then one or two stronger models, and compare them on the test set.
5. Choose evaluation metrics that suit the class balance, such as precision, recall and ROC-AUC, rather than accuracy alone.
6. Add the results, a comparison table and a short discussion to this report.